SECP 2753 Data Mining
Review Questions - MODULE 1A - INTRODUCTION TO DATA MINING

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Instruction:

Answer ALL the Questions and submit through elearning latest at 5 pm (Wednesday 19 March 2025)

1. What is Data Mining, and why is it important?

   Data Mining is a process of extracting or "mining" knowledge from large amounts of data. It is important because it helps in decision-making across various fields.

2. What are the key steps in the Knowledge Discovery in Databases (KDD) process?

   Cleaning and Integration, Selection and Transformation, Data Mining, Evaluation and Presentation.

3. Explain the general architecture of a data mining system

   Data Mining system works with several key components. The data warehouse serves as a data source while its server retrieves data for data mining process. The knowledge base guide the mining process. Then, the data mining engine consists a set of functional modules for tasks. For pattern evaluation module employs interestingness measure and interacts with data mining. Lastly, the graphical user interface used to communicates between users and data mining system.

4. What are some types of data that can be mined?

   Relational databases, data warehouses, transactional databases, advanced databases and the web and social media.

5. How is Data Mining related to Machine Learning, Artificial Intelligence, and Statistics?

   Machine Learning is a combination of statistic and AI, Artificial Intelligence uses heuristic-based approaches for learning patterns and Statistics provides analysis for study data. The relationship between them is there is link of statistic and AI to form Machine Learning and data mining uses Machine Learning patterns to discovering large datasets.

6. What are the two main types of Machine Learning

   Supervised Learning and Unsupervised Learning.

7. Provide real-world applications of Data Mining.

   Business like market analysis to understand the trend of purchasing and Healthcare like analysing disease and provide treatment predictions.

8. How does Data Mining differ from simple data analysis and information retrieval?

   Data Mining is basic analysis by discovering hidden patterns and relationships in large datasets. Data Analysis focus on summarizing and describing data while Information Retrieval is finding relevant documents from large text collection.